

1 **Waveform Identity and Reproducible Source-Grouped Evaluation of Seismic Phase Pickers**

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19 **Abstract**

20 Reliable comparison of seismic phase pickers requires traceable data identities and inspectable
21 relationships between training observations and derived predictions. We present an executable
22 evaluation methodology linking waveform fingerprints, earthquake-source groups, out-of-fold
23 feature construction, model provenance and numerical verification. An exhaustive audit of the
24 local STanford EArthquake Dataset archive uniquely matches 2,234 retained waveforms to
25 1,477 sources. A tested export profile also regenerates all 2,234 retained CSV byte hashes. A
26 fixed source-disjoint split contains 330 test records from 221 sources. Fresh replay loads all
27 106 checkpoints, reproduces 18,528 out-of-fold and 1,980 test feature tensors, and recovers 70
28 test prediction sets; independent calculations reproduce 1,260 metric rows. Synthetic examples
29 demonstrate rejection of invalid source membership, augmentation lineage, feature order and
30 artifact bytes. Temporal stacking and a PhaseNet-style baseline provide a case study of
31 component access. On this fixed split, stacking achieves accepted-pick F1 scores within 100
32 ms of 0.8268/0.5536 for compressional/shear arrivals with three components and
33 0.8134/0.2686 with vertical input alone. The shear-phase difference is 0.2850, with a
34 conditional source-cluster 95% interval of 0.2293–0.3418. The baseline has a supported
35 compressional-phase advantage; primary shear-phase method-comparison intervals include
36 zero. The contribution is an executable connection between identity, data lineage and
37 evaluation, rather than a new stacking algorithm. Historical development exposure prevents an
38 independent-confirmation claim. One fixed split, limited unseen-station observations and
39 unresolved historical selection decisions constrain predictive interpretation.

40 **Keywords: seismic phase picking; waveform identity; data provenance; source-grouped**
41 **evaluation; reproducibility; temporal stacking**

42 **1. Introduction**

43 Seismic phase-picking results are only interpretable when the evaluated waveforms, their
44 earthquake sources and their roles in model development can be traced. Arrival times of
45 compressional (P) and shear (S) waves support earthquake location and subsurface analysis,

46 but a reproducible performance number also depends on record identity, partition membership
47 and decoding rules. Earth science data identification therefore concerns both locating an
48 archived object and determining whether differently represented objects contain the same
49 scientific information (Duerr et al. 2011). In a learned processing pipeline, these identifiers
50 must remain connected to augmentation, fitted models and derived predictions.

51 Classical onset detectors use energy changes or change-point criteria (Allen 1978; Baer and
52 Kradolfer 1987; Maeda 1985; Sleeman and van Eck 1999), while adaptive characteristic
53 functions extend automatic P-wave estimation (Cheng et al. 2024). Neural pickers learn
54 probability sequences from waveforms (Zhu and Beroza 2019; Mousavi et al. 2020).
55 Physically driven feature engineering offers another route for detection and phase picking in
56 seismo-volcanic data (Vargas-Zamudio et al. 2026). Shared benchmarks and software make the
57 dependence of performance on data distribution and evaluation protocol explicit (Münchmeyer
58 et al. 2022; Woollam et al. 2022). These approaches motivate inspecting evaluation inputs and
59 outputs alongside model architecture.

60 Validation design can change the interpretation of seismic machine-learning results. Jover-
61 Alfaro et al. (2026) examined random, temporal and cross-location evaluation in earthquake
62 forecasting. Their forecasting task differs from arrival-time picking, and their findings are not
63 evidence of leakage in the pickers considered here. The relevant informatics problem is how to
64 make the dependencies between observations, training and evaluation inspectable. For phase
65 picking, repeated recordings of the same earthquake require a source identifier; chronological
66 ordering alone does not express every such dependency. Source grouping also does not remove
67 station or regional dependence, or erase earlier development exposure.

68 Leakage in scientific machine learning can arise from dependencies between observations and
69 research decisions, not only direct inclusion of test labels in fitting (Kapoor and Narayanan
70 2023). Cross-validation must respect the dependence structure relevant to the intended
71 prediction task (Roberts et al. 2017). Repeated optimization of a finite validation criterion can

72 itself produce selection bias (Cawley and Talbot 2010), which motivates explicit disclosure of
73 outer-validation reuse here.

74 Scientific provenance describes entities, activities and their derivations (Moreau and Missier
75 2013), while FAIR principles emphasize findability, accessibility, interoperability and reuse
76 (Wilkinson et al. 2016). SeisBench provides dataset and model interfaces for seismology
77 (Woollam et al. 2022). Our narrower contribution is to connect recovered exported-waveform
78 identities to group memberships, feature producers and frozen outputs in a tested workflow. We
79 do not claim that metadata toolkits or general provenance standards lack these capabilities; we
80 document the specific invariants and observed failures in this case.

81 Stacked generalization trains a second-level learner from base-model predictions (Wolpert
82 1992). Temporal stacking retains probability sequences, exposing disagreement in timing and
83 confidence to a fusion model. It introduces additional data lineage: the meta-training inputs
84 depend on which observations fitted each base model. We implement source-grouped out-of-
85 fold (OOF) construction with separate inner validation for each held fold, a canonical 15-
86 channel interface, augmentation-parent checks, source-membership assertions and input/output
87 hashes. The final base and meta stages reuse outer validation for selection; this dependency is
88 documented rather than represented as independent selection.

89 Recent fusion approaches combine complementary representations. TwoStream-EQT uses
90 time-domain and time-frequency inputs and evaluates STEAD, DiTing and Geysers data (Peng
91 et al. 2025). SeismoDual combines waveforms and spectrograms for P-phase picking across
92 STEAD, INSTANCE and CWA (Tang and Chen 2026). Here, fusion uses temporal outputs of
93 heterogeneous pickers together with waveform context. We compare it with a PhaseNet-style
94 MATLAB baseline under matched data-access and decoding rules. The study does not infer
95 missing controls in those published approaches, establish superiority over their reported scores,
96 or claim a new neural architecture.

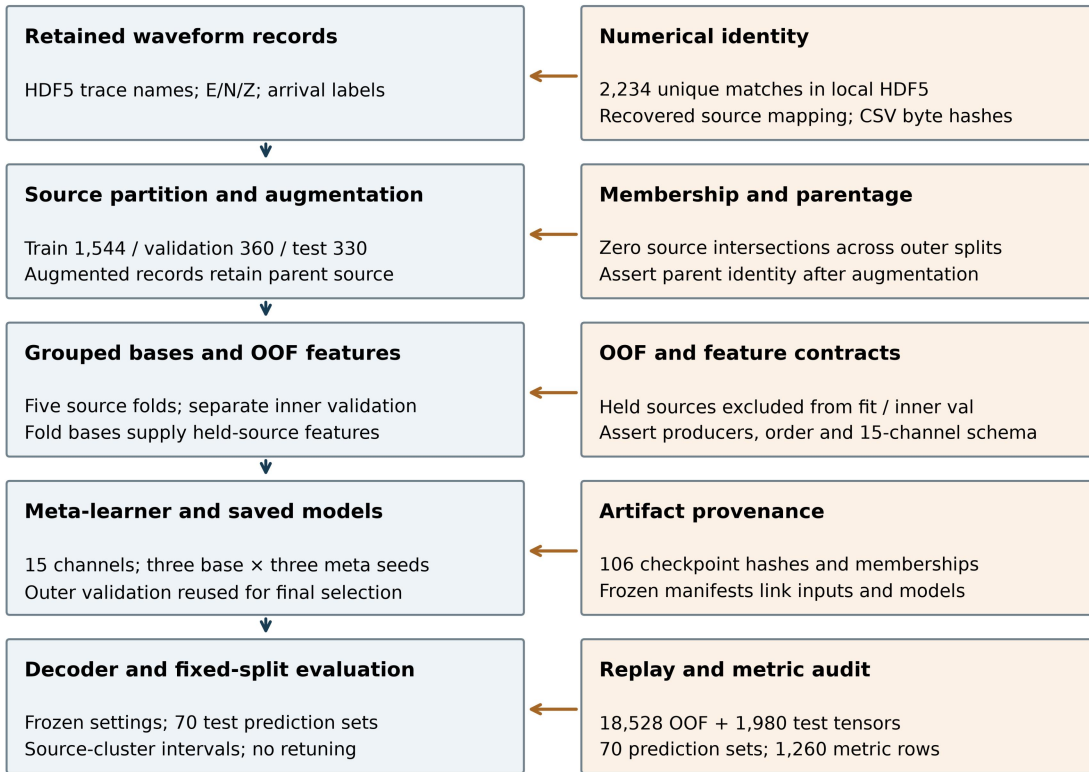
97 The study addresses three questions: RQ1, can retained exports be linked uniquely to
98 waveform records and earthquake sources in the local archive; RQ2, can those identities be

99 followed through grouped fitting, out-of-fold features and fixed-model replay; and RQ3, what
100 component-access and method comparisons are supported on the fixed split? The
101 methodological contribution is a case-tested integration of identity resolution, executable
102 membership and feature contracts, and numerical verification. These established techniques are
103 connected across the pipeline rather than proposed as new algorithms. Successful replay is
104 distinguished from independent predictive confirmation.

105 Figure 1 shows data and model flow alongside the checks applied at each stage. Results first
106 answer RQ1 and RQ2, then report the expanded base/meta-seed comparison for RQ3. Base-42
107 ablations and detailed error diagnostics are secondary analyses in Online Resource 1. Three
108 base initializations crossed with three meta initializations characterize variation on one fixed
109 source split; they do not represent nine independent pipelines.

DATA AND MODEL FLOW

VERIFICATION AND EVIDENCE



Historical development exposure remains: these checks do not establish independent confirmation.

Solid vertical arrows: computational flow. Horizontal arrows: checks applied to each stage.

110

111 Fig. 1. Data/model flow and verification layer. Checks connect numerical waveform identity,
 112 source partitions, augmentation parentage, OOF membership, feature order, saved artifacts and
 113 reported metrics. Outer validation is reused for final base selection, meta selection and
 114 ensemble weighting. Historical development exposure remains outside the exclusions
 115 established by the corrected run. Evidence types and enforcement scope are distinguished in
 116 Table 3.

117 2. Data and dataset curation

118 2.1. Source dataset

119 We used the Stanford EArthquake Dataset (STEAD; Mousavi et al. 2019). The retained
 120 population comprises 2,234 previously exported local-earthquake waveform records whose
 121 identities were recovered by matching their numerical content to the upstream HDF5 data. It
 122 contains 1,477 earthquake sources. Recovered metadata show magnitude 2.5–5.9, source
 123 distance 0–49.99 km, and minimum component SNR −1.70–103.60 dB (median 44.70 dB; Fig.

2; Table 1). The original extraction index and selection script were unavailable; these are measured population characteristics, not a reconstructed selection rule. Only 484 records have distance ≤ 15 km and 2,172 have minimum component SNR ≥ 10 dB. We therefore do not characterize the entire subset by those previously stated filters.

2.2. Waveform representation and record format

Each event–station observation contains 6000 samples at 100 Hz (60 s), with ordered E, N, and Z components and reference arrivals in seconds. The waveform representation is given in equation 1; equation 2 converts an arrival time to a zero-based sample index.

$$\mathbf{X}_i = \begin{bmatrix} E_i(t_1) & N_i(t_1) & Z_i(t_1) \\ E_i(t_2) & N_i(t_2) & Z_i(t_2) \\ \vdots & \vdots & \vdots \\ E_i(t_{6000}) & N_i(t_{6000}) & Z_i(t_{6000}) \end{bmatrix} \quad (1)$$

where $\mathbf{X}_i \in \mathbb{R}^{6000 \times 3}$ is the three-component waveform for record i . The associated ground-truth arrival times are denoted by t_i^P and t_i^S . Sample indices were derived only when required by the processing functions (equation 2):

$$n_i^P = \text{round}(t_i^P f_s), \quad n_i^S = \text{round}(t_i^S f_s) \quad (2)$$

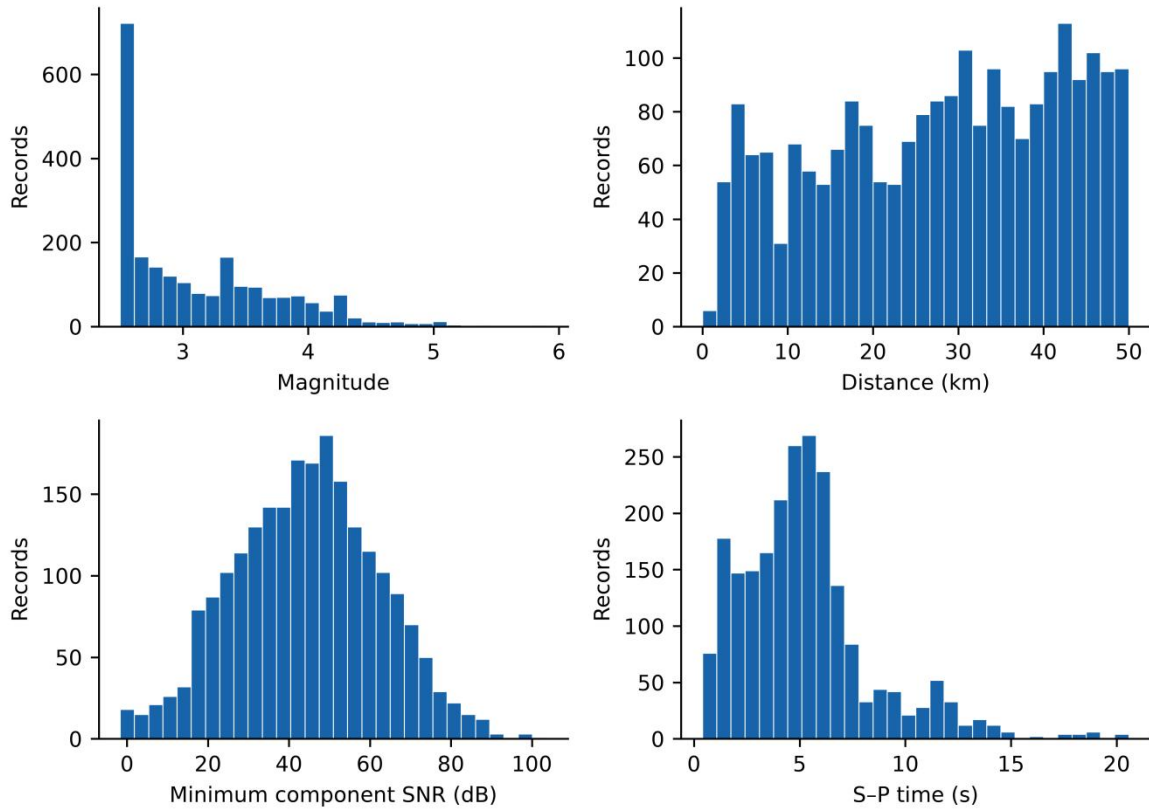
Here $f_s = 100$ Hz. A zero-based sample n maps to MATLAB array index $n+1$; an array index j maps back to time $(j-1)/f_s$. CSV arrival times in seconds remain authoritative, and evaluation does not replace them with rounded metadata indices.

Original STEAD source_id values recovered from waveform matches define grouping; event IDs identify individual exported records. All 2,234 CSV labels equal floor(upstream arrival sample)/100 for both phases. Fractional upstream samples differ from the retained CSV labels for 94 P and 124 S arrivals, by at most 9.01 ms. We preserve the historical CSV labels consistently across methods rather than silently replacing their rounding convention.

2.3. Metadata integration and identifier recovery

We matched $6,000 \times 3$ CSV waveforms to STEAD using SHA-256 hashes of little-endian float32 values in column-major E/N/Z order. Exhaustive hashing of all 1,265,657 local HDF5 waveform datasets found exactly one trace for each of the 2,234 distinct selected fingerprints,

149 without conflicting source assignments. Upstream metadata agreed throughout. The mapping
150 and partitions were frozen before training of the corrected source-verified run, termed v2;
151 exhaustive uniqueness verification followed training. This establishes exact matches in this
152 archive version, not near-duplicate absence or catalogue identity accuracy.



153

154 Fig. 2. Characteristics of the curated STEAD-derived dataset (N=2234 records): magnitude,
155 epicentral distance, minimum component SNR, and S–P time. Each histogram uses 30 bins.

156 Table 1. Summary statistics of the curated STEAD-derived dataset used in this study.

Item	Value
Records / recovered sources	2234 / 1477
Sampling / duration	100 Hz / 60 s
Components	E, N, Z
Magnitude range; median	2.5–5.9; 2.9
Distance range; median (km)	0–49.99; 29.74
Minimum SNR range; median (dB)	–1.70–103.60; 44.70

157
 158 Identity correction and rerun. The original ordinal metadata join misidentified waveforms:
 159 151/335 historical test records shared physical sources with training. We corrected identities
 160 and reran all 106 fits. Current test records previously belonged to training (233), validation (46)
 161 or testing (51); 196/221 current test sources had historical training/validation membership.
 162 Thus source disjointness in v2 does not establish independent confirmation. Because
 163 population allocation and training also changed, old-to-new scores cannot isolate leakage-
 164 induced inflation (Online Resource 1, Tables S22–S23).

165 **2.4. Waveform quality control**

166 The corrected loader accepted all 2,234 records without missing files or read failures. Input
 167 checks cover waveform dimensions, finite values, the 100 Hz time grid and in-window ordered
 168 arrival labels. Numerical hashes connect these inputs to the recovered STEAD traces. No new
 169 filtering or label correction was applied after the verified population and source split were
 170 frozen.

171 **2.5. Signal conditioning**

172 Conditioning removes the component mean, detrends linearly, and applies a 1–45 Hz
 173 Butterworth band-pass filter using MATLAB butter with order argument 4 followed by filtfilt.
 174 Each component is normalized as $(x - \text{mean}(x)) / (\text{std}(x) + 10^{-10})$, so an all-zero component
 175 remains zero. Explicit clipping was disabled. The same deterministic transforms were used in
 176 all subsets (equation 3). Supplementary Implementation Details specify the enhanced features
 177 and edge handling.

178 For an input component $x_c(t)$, where $c \in \{E, N, Z\}$, the conditioning operation can be
 179 summarized as (equation 3):

$$180 \quad \tilde{x}_c(t) = \mathcal{N}\{\mathcal{B}_{1-45\text{Hz}}[\mathcal{D}(x_c(t) - \bar{x}_c)]\} \quad (3)$$

181 where $\mathcal{D}$ denotes detrending, $\mathcal{B}_{1-45\text{Hz}}$ denotes band-pass filtering, and $\mathcal{N}$ denotes component
 182 normalization. The same deterministic conditioning parameters were applied to training,

validation, and test records. Data augmentation was not part of dataset curation and was applied only after splitting, exclusively to training data.

Base CNN/TCN inputs contain eight channels: conditioned E/N/Z, their three envelopes, short-term Z energy, and a Z characteristic function. The final three waveform-context channels of the meta tensor retain conditioned E/N/Z; they are distinct from the eight-channel base representation.

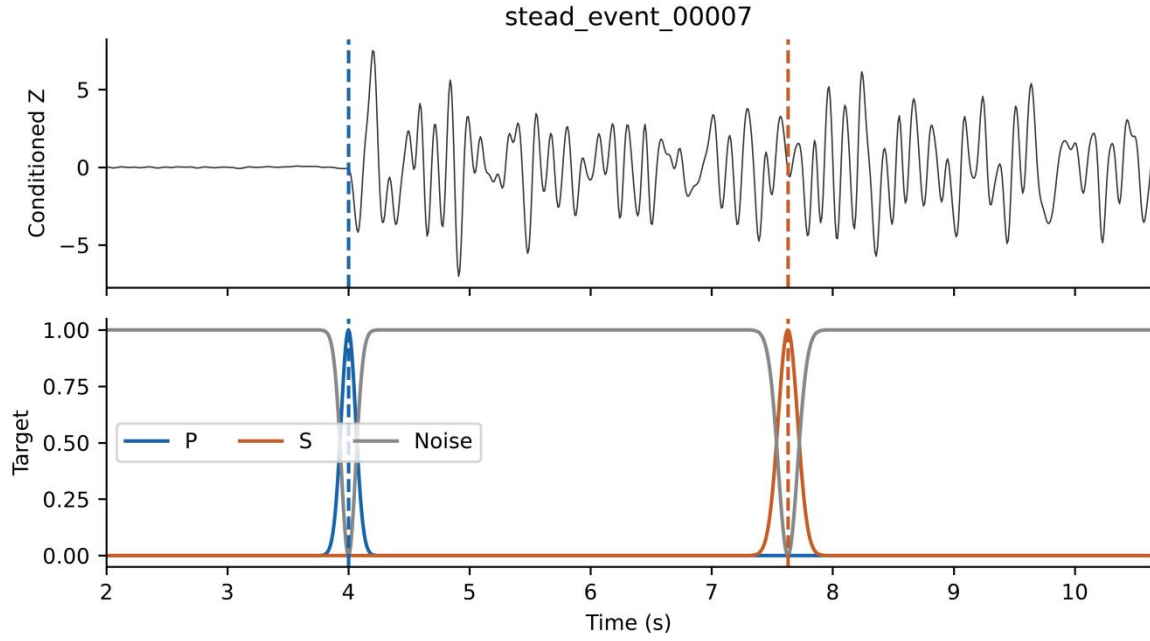
2.6. Archive reference labels and target construction

Reference arrivals became Gaussian phase targets (equation 4; Zhu and Beroza 2019; Mousavi et al. 2020). Both arrivals have manual metadata status in 1,224/1,544 training, 306/360 validation and 287/330 test records. The other 417 records include automatic/autopicker status. Manual-only sensitivity restricts evaluation without retraining or independently validating annotations (Online Resource 1, Tables S24–S25). Reference labels come from archive metadata, not independent expert re-picking in this study.

For phase $k \in P, S$, the target at time t was defined as (equation 4):

$$y_k(t) = \exp \left[-\frac{(t-t_k)^2}{2\sigma_k^2} \right], \quad (4)$$

Targets were truncated at ± 4 standard deviations, with P/S widths of 0.06/0.08 s. The noise target is $\max(0, 1 - \max(y_P, y_S))$; targets are not additionally normalized to sum to one where phase tails overlap (Fig. 3). Neural predictions, in contrast, use softmax. Target mass is $1 + \min(y_P, y_S)$. Overlapping masks occur in only 4 training, 0 validation, and 1 test records (Online Resource 1, Table S14). The implemented loss and its local weighting interpretation are given in Supplementary Implementation Details.



204

205 Fig. 3. Gaussian probability label formulation. (a) Vertical (Z) seismic component with true P-
 206 and S-wave arrival markers. (b) Gaussian probability masks used as training targets for the I-
 207 CNN meta-learner. The P-wave mask (blue) and S-wave mask (red) are centred on their
 208 respective true arrival times with standard deviations σ_P and σ_S (in samples). The
 209 noise/background mask (grey) equals $\max(0, 1 - \max(y_P, y_S))$, the continuous complement of
 210 the larger phase mask.

211 2.7. Source-level partitioning

212 Unique recovered source IDs were deterministically ranked using SHA-256 stage/seed/source
 213 keys and allocated approximately 70%, 15% and 15% to train, validation and test. All records
 214 of one earthquake stay together (Table 2). Training contains 1,544 records from 1,034 sources,
 215 validation 360 records from 222 sources, and test 330 records from 221 sources. Pairwise
 216 source intersections are zero; station and regional overlap were not excluded. The same frozen
 217 outer split and five OOF assignments are used for both input modes and every base seed.
 218 Subset distributions appear in Online Resource 1, Fig. S2.

219 Test metadata contain 64 network–receiver keys across 23 networks; PB contributes 99/330
 220 records. Fifty-five test keys also occur in training, covering 318 records. Nine keys absent from

221 training and validation contain only 12 records from ten sources. These are retrospective strata,
 222 not a station-held-out design; keys do not distinguish station epochs (Online Resource 1,
 223 Tables S12 and S26).

224 Table 2. Source-level partitioning of the final curated dataset showing the number of unique
 225 earthquake sources and waveform records in each subset.

Subset	Sources	Records
Training	1034	1544
Validation	222	360
Test	221	330
Total	1477	2234

226 **2.8. Leakage controls after dataset partitioning**

227 Augmentation was restricted to training records: one augmented copy per original used
 228 additive noise at 20 dB target SNR, amplitude scaling in [0.5, 2.0], and horizontal-channel
 229 dropout with probability 0.20 (E, N, or both). Time shifting and polarity reversal were disabled.
 230 Validation and test records were not augmented. Source assignments were preserved
 231 throughout processing.

232 **3. Methods**

233 **3.1. Workflow objects and executable contracts**

234 A record is identified by its local record key, recovered STEAD trace name, source identifier
 235 and numerical waveform fingerprint. An augmented record also stores a parent relationship. A
 236 fold model has fitting and inner-validation source sets; a final model has recorded fitting and
 237 selection memberships. A feature tensor adds ordered producer/channel identities, and an
 238 artifact adds a byte hash. These objects are distinct: matching bytes alone does not validate a
 239 source key, and source disjointness alone does not remove earlier development exposure.

240 Table 3. Workflow contracts and observed evidence. Checks are stated as invariants; the
 241 evidence column distinguishes full-data audits from synthetic demonstrations.

Contract	Check and stage	Evidence and scope
Record identity	A retained record maps to one local HDF5 trace with the same numerical fingerprint.	2,234 unique matches in the exhaustive local-archive scan; not a near-duplicate test.

Contract	Check and stage	Evidence and scope
Outer source exclusion	Source sets for train, validation and test have pairwise empty intersections. Check frozen protocol before fitting.	Zero intersections across 1,034 / 222 / 221 sources; historical exposure remains.
Augmentation parent	Child source equals parent source and child points to its parent record.	Training augmentation preserves membership; synthetic changed-parent case rejected. No claim of auditing every historical augmentation.
OOF exclusion	For fold f , held-source set H_f has empty intersection with Fit_f and $InnerVal_f$. Audit memberships before interpreting OOF outputs.	Six mode/base membership manifests $\times$ 1,544 records; synthetic contaminated-fold case rejected.
Feature interface	15 ordered channels from named producers; held originals and copies use their fold-specific bases.	Schema assertions and swapped-feature negative control; real OOF/test reconstruction checked separately.
Checkpoint and bytes	Recorded fit/validation sources exclude current test sources; artifact hashes match frozen manifests.	106 checkpoints audited; 106 model and 89 code-file hashes unchanged across recorded replay.
Numerical output	Replayed features/picks and recomputed metrics agree with saved references under stated comparisons.	18,528 OOF + 1,980 test tensors; 70 prediction sets; 1,260 metric rows. Selection history is not replayed.

242 The core invariants are pairwise exclusion of outer source sets, equality of child and parent
 243 source identity, and disjointness of each held fold from both fitting and inner-validation
 244 sources. The implementation checks these relationships using membership manifests and
 245 assertions. The synthetic demonstration exercises the same types of failure, but is not a
 246 substitute for audits of the fitted study artifacts. Table 3 records the scope of each evidence
 247 item.

248 Experiments used MATLAB R2024a with fixed source partitions. The corrected 40-fit main
 249 experiment used CNN/TCN base seed 42 and meta seeds 42, 43, and 44. An additional 66 fits
 250 repeated the complete base/OOF/meta workflow with base seeds 43 and 44 and trained six
 251 standalone PhaseNet-style baselines. Each base seed was crossed with all three meta seeds in
 252 both modes. OOF fold seed 42, inner split seeds 101–105, fold augmentation seeds 1001–1005,
 253 and final/OOF augmentation seed 4242 remained fixed. Thus initialization and training-order
 254 variability is examined conditional on the same partition and augmentation protocol; meta fits
 255 sharing a base are not independent full-pipeline replicates. Online Resource 1, Table S9

enumerates the 40 main invocations; Table S18 accounts for the full 106 fits. Full3C denotes access to all three waveform components; Z-only retains vertical input with E/N set to zero. I-CNN denotes the integrated convolutional meta-learner used for temporal stacking.

3.2. Base score generators and grouped fitting

Conditioning follows equation 3. STA/LTA and AIC supply deterministic probability-like scores rather than calibrated class probabilities (Allen 1978; Maeda 1985). Gaussian curves are centred on the global normalized STA/LTA peak or the AIC minimum. P uses Z; S uses the E/N resultant only when both horizontal components are nonzero, otherwise it falls back to Z. The AIC centre search is restricted to the first 10 s. Supplementary Implementation Details give the exact windows, normalization, fallback, and score construction. Learned bases emit P/S/noise softmax probabilities (equation 5). These classical channels are retained historical score generators, not optimized conventional-picker benchmarks; their restricted AIC centre search is evaluated as implemented.

The CNN has two convolutional blocks with 32/64 filters, kernel 7, batch normalization, ReLU, and dropout 0.30; its receptive field is 13 samples. TCN has four 32-filter blocks with kernel 3, dilations 1/2/4/8, dropout 0.20, and no residual connections; its receptive field is 31 samples (Bai et al. 2018). Both process the eight-channel representation at 100 Hz and finish with a 1×1 convolution and softmax. Base training used Adam at 10^{-3} , batch size 16, at most 30 epochs, and validation patience of seven evaluations (Online Resource 1, Table S28).

$$P_P(t) + P_S(t) + P_N(t) = 1 \quad (5)$$

Five source-grouped OOF folds were formed within outer training. For each held fold, the other four were split by source into fitting and inner validation, reserving approximately 15% for checkpoint selection. Held sources entered neither subset. Those fold models supplied OOF features for held originals and their augmented copies. Final bases fitted weights using augmented outer training only; outer validation selected their checkpoints. Final bases then supplied outer-validation and test meta-features. Meta weights fitted OOF training features; the

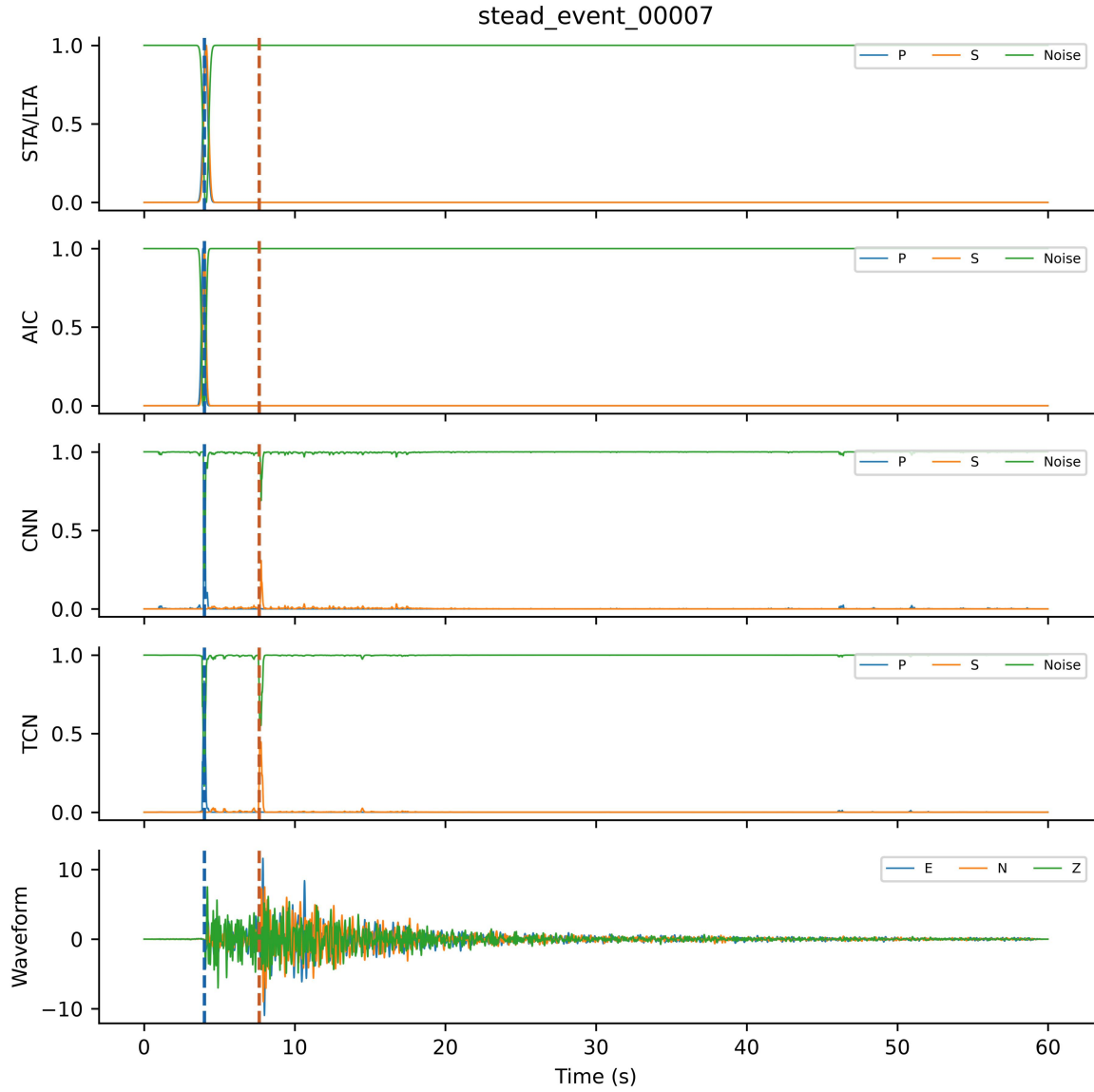
282 same outer validation selected meta checkpoints. This is adaptive validation reuse across levels,
 283 not an independent nested validation estimate. Current test sources were excluded from v2
 284 fitting and checkpoint selection; this exclusion does not erase their historical roles before
 285 identity correction. Online Resource 1, Table S8 maps each role.

286 3.3. Feature interface and temporal models

287 The meta input is a 6000×15 tensor: twelve P/S/noise channels from four pickers, followed by
 288 conditioned E/N/Z (equation 6; Fig. 4). Channel count and order are asserted. Online Resource
 289 1, Table S1 describes all held-out test tensors; Online Resource 1, Fig. S5 illustrates one record.
 290 Table 4. Canonical temporal feature interface. Channel indices remain fixed in both input
 291 modes.

Channels	Producer and values	Z-only behavior
1–3	STA/LTA: P, S, noise scores	P and S coincide under vertical fallback
4–6	AIC: P, S, noise scores	P and S coincide under vertical fallback
7–9	CNN: P, S, noise probabilities	Mode-specific fitted base
10–12	TCN: P, S, noise probabilities	Mode-specific fitted base
13–15	Conditioned E, N, Z waveforms	E/N zero; Z retained

292 $\mathbf{X}_{\text{meta}} \in \mathbb{R}^{6000 \times 15}$ (6)



293

294 Fig. 4. Final 15-channel tensor for held-out `stead_event_00007`. Four groups contain P/S/noise
 295 from STA/LTA, AIC, CNN and TCN; the last group contains conditioned E/N/Z. Dashed lines
 296 mark true arrivals. These are current final-base test features, not OOF training features.

297 I-CNN has four sequential kernel-5 blocks with filters 64/128/64/64, dilations 1/1/2/4, batch
 298 normalization, ReLU, and dropout 0.30. With $h_0 = \text{Xmeta}$, equations 7–8 define the blocks and
 299 head. Its receptive field is 33 samples (330 ms); architectural capacity alone does not establish
 300 an empirical benefit. The short-context stacker is a hierarchical testbed for lineage checks; no
 301 competitive-architecture claim is made.

302 For blocks $i = 1, \dots, 4$, the forward operation is:

$$303 \quad h_i = \text{Dropout}(\text{ReLU}(\text{BN}_i(W_i *_{d_i} h_{i-1} + b_i))) \quad (7)$$

304 Here $d = (1, 1, 2, 4)$, the convolution uses dilation d_i and same padding, and dropout probability
305 is 0.30 during training only. The 1×1 head returns three probabilities at each time:

$$306 \quad p(t) = \text{softmax}(W_{\text{head}} h_4(t) + b_{\text{head}}) \quad (8)$$

307 The I-CNN used Adam at 5×10^{-4} , batch size 16, at most 50 epochs, and weighted cross-entropy
308 with P/S/noise weights 1.5/2.5/0.5. All neural models used $L2 = 10^{-4}$, learning-rate halving
309 every 20 epochs, validation once per epoch, and the best-validation-loss checkpoint. Meta
310 patience was ten evaluations. Histories appear in Online Resource 1, Fig. S1.

311 Training-only augmentation is specified in Section 2.8 and Online Resource 1, Table S28.
312 Adam (Kingma and Ba 2015), batch normalization (Ioffe and Szegedy 2015), and dropout
313 (Srivastava et al. 2014) were used for neural models. Online Resource 1, Table S4 records the
314 final configuration, including disabled augmentation operations and seed scope. Horizontal
315 dropout is triggered with total probability 0.20 per augmented copy; E-only, N-only, and both
316 are then selected with equal probability. Losing either horizontal component activates the Z
317 fallback in both classical bases.

318 For Z-only, E and N were zeroed before preprocessing and again after augmentation.
319 Population, source partitions, labels, channel layout, architecture settings, and evaluation were
320 unchanged; mode-specific bases and meta-learners were fitted. The comparison removes
321 horizontal information throughout the pipeline, rather than masking only waveform-context
322 channels after full-component base inference. For the audited base-seed cache, all 3,778
323 available Z-only meta tensors were finite with zero E/N channels; classical P/S score channels
324 were identical under the specified fallback (Online Resource 1, Table S13).

325 The standalone comparator, termed PhaseNetMatched, is a PhaseNet-style MATLAB
326 adaptation trained from scratch (Zhu and Beroza 2019; Woollam et al. 2022). It has five

encoder levels with 8/16/32/64/128 filters, kernel 7, stride-4 downsampling, batch normalization, ReLU, transposed convolutions, and skip concatenation. MATLAB same padding and centre-cropped skips support 6,000-sample records. Padding, initialization, convolution biases, and batch-normalization defaults differ from the reference implementation. The input is the same conditioned E/N/Z waveform used in meta channels 13–15, without a second normalization; E/N remain zero in Z-only. Labels, augmentation, weighted loss, outer split, validation selection, decoder, and scoring are shared. Adam uses learning rate 5×10^{-4} , batch size 2, at most 50 epochs, patience 10, L2= 10^{-4} , and learning-rate halving every 20 epochs. Three seeds (42–44) are fitted per mode. Data access is matched; feature inputs, minibatch statistics, and compute budgets are not equivalent. Online Resource 1, Table S18 accounts for all additional fits.

3.4. Arrival decoder

Let $G = \{0, 0.01, \dots, 59.99\}$ s. Set $DP = G$. The P candidate maximizes pP over G . If it is accepted, the S search domain is restricted relative to that candidate; otherwise the S domain is empty:

$$D_s = \begin{cases} \{t \in G : 0.1 \leq t - t_p \leq 30\}, & \text{if } P \text{ is accepted} \\ \emptyset, & \text{otherwise} \end{cases} \quad (9)$$

DS is empty when P is not accepted. For each nonempty phase domain D_k , $\hat{t}_k$ maximizes p_k on D_k . Ties use the earliest sample, matching MATLAB max. The peak-to-mean quality score is computed on that phase domain:

$$q_k = \frac{p_k(\hat{t}_k)}{\text{mean}\{p_k(t) \mid t \in D_k\} + 10^{-6}} \quad (10)$$

A candidate is detected when $\text{peak} \geq 0.30$ and $\text{quality} \geq 3.0$, otherwise uncertain when $\text{peak} \geq 0.15$, and not detected below 0.15. Primary acceptance includes both detected and uncertain; strict reporting includes detected only.

Return $tS = \hat{t}S$ for accepted S and NaN otherwise; an empty DS returns no pick. The S–P rule constrains ordering rather than learning a physical objective. Detected-only scoring retains the accepted-P gate. Peak 0.30, quality 3 and the 0.1–30 s window occur in July and September configurations (Online Resource 1, Table S20), but their earliest calibration data remain unknown. No validation-only calibration is asserted. Before sensitivity calculations, we specified a 27-setting grid: peak {0.20, 0.30, 0.40}, tied P/S quality {2, 3, 4}, and maximum S–P {20, 30, 40} s. Minimum separation remains 0.1 s and the uncertain boundary half the peak threshold. All 18 full stackers and six baseline fits retain the original primary setting. Window changes also alter the S quality denominator; accepted scoring is invariant to quality threshold.

3.5. Evaluation and dependency-aware summaries

Evaluation used 330 test records from 221 sources and tolerances of ± 50 , ± 100 , and ± 200 ms. The primary policy accepted finite detected or uncertain picks. Accepted picks within tolerance were true positives; accepted picks outside tolerance contributed both a false positive and a false negative. Missing picks contributed false negatives. Thus $FP = N_{\text{accepted}} - TP$ and $FN = N - TP$. Detected-only sensitivity and seed-42 confusion counts are reported in Online Resource 1, Tables S10 and S7. The primary policy retains lower-confidence candidates to expose the coverage–timing tradeoff; accepted coverage must not be interpreted as accurate detection. Every test record contains P and S reference arrivals; there are no noise-only records. Here false positives count mistimed accepted picks, and precision is not a continuous-stream false-alarm measure.

The conventional F1-score is (equation 11):

$$F1 = \frac{2TP}{2TP+FP+FN} \quad (11)$$

MAE was calculated over all accepted arrivals, including uncertain picks and errors outside the matching tolerance (equation 12):

$$MAE = \frac{1}{N_d} \sum_{i=1}^{N_d} |t_i - \hat{t}_i|, \quad (12)$$

In equation 12, N_d counts accepted finite picks; coverage N_d/N differs from recall and detected-only coverage. Expanded stacking summaries average meta seeds within each base and then the three base means. Between-base SD uses those three means, not nine independent replicates; baseline summaries use three fits. Paired intervals use 10,000 NumPy default_rng source-cluster draws (seed 42), retaining every record of each sampled source. Per-fit F1 is recomputed before averaging, with common draws across comparisons. Intervals condition on fixed models and split; they exclude training, partition and separate station-level uncertainty. Subset diagnostics resample their stated sources. Intervals are exploratory and pointwise, without family-wise or false-discovery-rate control. No configuration is selected from the threshold grid. Base-42 summaries retain their three-meta-seed scope.

3.6. Verification and reproduction levels

The verified workflow checks source disjointness, augmentation parentage, OOF exclusion, the 15-channel feature schema and file hashes. A retained synthetic example illustrates invalid source overlap, changed parent identity, OOF contamination, swapped feature order and altered bytes; it is an implementation demonstration rather than evidence of predictive accuracy. Fresh inference replay and an independent numerical audit check all current prediction identities, reference labels, source membership and reported metrics. Online Resource 1, Tables S19 and S21 separate current-run evidence from historical software demonstrations.

Reuse follows three explicit input/output steps. First, retained waveform files are associated with source identifiers and a frozen partition manifest; source membership and augmentation parentage are checked before model fitting or feature assembly. Second, fold-specific predictions are assigned to the

canonical feature schema, with each OOF record excluded from the associated base fit. Third, replayed predictions are evaluated under saved decoder settings and compared with frozen output identities and metrics. The checks establish these specified invariants; they do not prove that the grouping key is scientifically correct for every new application.

Table 5 separates the levels of reproduction. The synthetic checks and metric verifier require no external waveforms. Fresh model replay additionally requires hash-matching CSVs, compatible MATLAB toolboxes and the saved checkpoints. A new deterministic HDF5-to-CSV reconstruction utility was checked against all retained file hashes (Section 4.2). It reconstructs the selected exports, not the missing historical filtering and selection procedure. For another application, the grouping key, record mapping, labels and decoder must be supplied and justified.

Table 5. Reproduction levels and their limits. The new reconstruction check is separate from the previously completed model replay.

Level and inputs	What can be checked	Boundary
A: package, Python	Hash integrity; five synthetic negative controls; 1,260 metric rows from supplied predictions	No waveform or MATLAB needed; does not rerun models
B: STEAD HDF5 + mapping + Python	Regenerate retained CSVs and compare frozen byte hashes; 2,234/2,234 matched	Tested serialization and local archive; original selection decisions not recovered
C: matching CSVs + checkpoints + MATLAB	Fresh OOF/test features and test predictions under saved settings	106 checkpoints; recorded GPU replay. Validation tensors and selection process not rebuilt
D: retraining or independent confirmation	Not established by A–C	No bitwise retraining claim; no new unexposed population tested

Fresh GPU replay loaded all 106 checkpoints and rebuilt inputs from waveform CSVs. Across six mode/base combinations, 18,528 original/augmented OOF tensors and 1,980 test meta tensors had maximum absolute difference 0 from their caches (acceptance tolerance 10^{-6}). All 70 prediction sets $\times$ 330 records reproduced constrained and unconstrained pick times and statuses without mismatches (time tolerance $1e-7$ s). Saved validation-derived ensemble weights were retained. This verifies fixed-artifact inference and feature reconstruction, not retraining determinism or independent predictive validation (Online Resource 1, Table S21). Outer-validation feature tensors were not rebuilt in this replay; saved ensemble weights and

420 selected checkpoints were reused. Thus replay does not revalidate the historical selection
421 decisions.

422 The recorded inference replay used MATLAB R2024a, Deep Learning Toolbox, Signal
423 Processing Toolbox and GPU support through Parallel Computing Toolbox, on an RTX 3060
424 Laptop GPU with 6 GiB VRAM and 32 GiB system RAM. These are observed resources, not
425 measured minimum requirements. Cross-device or CPU equivalence is not established. CSV
426 regeneration was tested with Python 3.12.14, NumPy 2.5.3, pandas 3.0.1 and h5py 3.16.0; it
427 verifies output hashes rather than assuming serialization is stable across versions.

428 Checkpoint inspection gives 16,611 trainable parameters for CNN, 10,467 for TCN, 108,355
429 for the full 15-channel I-CNN and 269,051 for PhaseNetMatched, identically in both
430 component modes. Counts include biases and batch-normalization scale/offset, but exclude
431 running statistics. The 106 recorded fit durations sum to approximately 16.73 h; this is
432 cumulative successful-fit wall-clock time, not end-to-end pipeline time. Preparation, OOF
433 generation, evaluation, checkpoint serialization and interrupted attempts are excluded. Per-
434 model durations and ablation counts appear in Online Resource 1, Table S31. A complete
435 replay wall-clock measurement is unavailable; a retained partial-run timer is explicitly scoped
436 in that resource.

437 **3.7. AI-assisted software development**

438 AI-assisted software development and computational workflow. ChatGPT/Codex (OpenAI;
439 accessed September 2026) assisted with software revision and debugging, workflow
440 implementation, computational execution support, and numerical-output checking. Assistance
441 included scripts for generating tables, data plots, and the workflow diagram. Figures were
442 produced programmatically from study data, model outputs, or the documented workflow,
443 without generative-image synthesis. AI tools did not generate seismic observations or reference
444 phase labels. The authors reviewed and verified code changes, executed analyses, numerical
445 results, and interpretations. Verification included source-membership assertions, numerical
446 consistency checks, and comparison of frozen outputs with reported tables. The verified v2
447 package and numerical audit identify the respective scripts, configurations, predictions and
448 provenance records.

4. Results

The evaluation contains 106 fits: 40 main/ablation fits, 60 additional base/OOF/meta fits and six PhaseNet-style fits. The principal presentation uses the expanded base/meta grid (Tables 6–7; Figs. 5–6). Base-42-only performance, ablations and error diagnostics are secondary analyses (Online Resource 1, Tables S5–S7 and S28–S30; Figs. S8–S14). These reporting priorities are retrospective, not preregistered hypotheses. The training code did not select checkpoints using current test scores.

4.1. Waveform identity and historical data roles

For RQ1, the exhaustive local-archive audit uniquely matched all 2,234 retained waveforms to 1,477 recovered sources. Current train, validation and test sources are disjoint, but historical exposure remains. Only 25 test sources (26 records) lack historical training/validation membership; all previously occurred in testing. No development-naive subset is established. Historical scores are not a valid source-held-out comparator, and the changed allocation and training prevent a controlled estimate of leakage inflation.

4.2. Feature reconstruction and numerical reproducibility

For RQ2, the identity mapping links 2,234 retained waveforms to checkpoint membership and fresh inference. OOF/test tensor differences were zero, with no pick/status mismatches across 70 prediction sets. Independent calculations reproduced 1,260 metric rows. The input audits cover 25,248 manifest entries across 2,336 unique input files; these are distinct from the 1,241 entries in the archived package-level manifest (Online Resource 1, Table S21).

Five deliberately invalid synthetic cases were rejected: source overlap, changed augmentation-parent identity, a held source included in an OOF fit, reordered feature channels and altered artifact bytes. These negative controls demonstrate the implemented checks, not robustness to every possible provenance error. Table 3 distinguishes synthetic demonstrations, retrospective real-data audits and replay evidence.

A deterministic serialization profile regenerated all 2,234 selected CSV byte streams directly from mapped HDF5 traces. Every source identifier, column-major little-endian float32 waveform fingerprint and complete CSV SHA-256 matched the frozen manifest, with zero errors. No original CSV content was used to generate these outputs. The check ran without training and without materializing the complete CSV collection. The utility can write only hash-matching files to a new output directory. This closes the tested retained-input reconstruction gap, while leaving the original selection/filtering history unresolved (Table 5; Online Resource 1, reconstruction verification).

4.3. Expanded component-access and method comparison

For RQ3, across three base-seed means, Full3C stacking F1@100 ms was 0.8268 ± 0.0069 for P and 0.5536 ± 0.0127 for S; Z-only values were 0.8134 ± 0.0073 and 0.2686 ± 0.0152 . Each base mean averages three meta fits. Full3C exceeded Z-only S F1 in all nine paired fits. Table 6 separates these summaries from the three-seed standalone baseline. Fig. 5 compares methods, and Fig. 6 separates individual meta fits from base-specific means. These repetitions characterize initialization variability on one source split, not uncertainty across partitions or regions.

Table 6. Expanded results at ± 100 ms. Stacking SD is across three base-seed means after averaging meta seeds; baseline SD is across three fits. MAE includes all accepted picks.

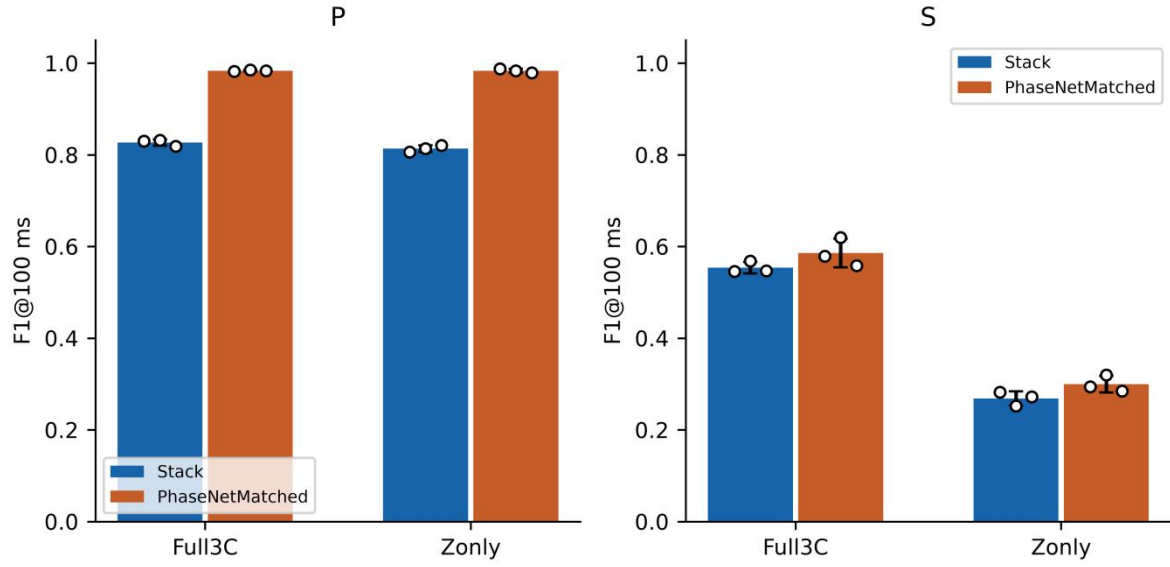
Method	Mode	P F1	S F1	P MAE ms	S MAE ms
Stack	Full3C	0.8268 ± 0.0069	0.5536 ± 0.0127	166.62	610.29
Stack	Z-only	0.8134 ± 0.0073	0.2686 ± 0.0152	178.30	1172.94
PhaseNetMatched	Full3C	0.9833 ± 0.0015	0.5857 ± 0.0313	34.47	170.12
PhaseNetMatched	Z-only	0.9833 ± 0.0045	0.2995 ± 0.0180	25.64	504.30

PhaseNetMatched achieved higher mean F1 and lower accepted-pick MAE in both modes. For P, stacker minus baseline $\Delta F1$ was -0.1565 in Full3C and -0.1699 in Z-only, with both conditional intervals below zero. For S, the corresponding differences were -0.0320 and -0.0309 , but both intervals included zero (Table 7). Full3C S accepted coverage was 98.01% for stacking and 99.39% for the baseline; Z-only values were 92.05% and 99.70%. These

497 results support the standalone P advantage. They neither establish an S advantage nor
 498 demonstrate S equivalence.

499 Table 7. Expanded mean $\Delta F1@100$ ms and paired source-cluster 95% intervals (10,000 draws).
 500 Differences follow the order shown. Intervals condition on fixed fits and one split and are not
 501 multiplicity-adjusted.

Comparison	Phase	Mean $\Delta F1$	95% interval
Stack Full3C minus Z-only	P	+0.0134	[-0.0016, 0.0295]
PhaseNetMatched Full3C minus Z-only	P	+0.0000	[-0.0102, 0.0112]
Stack minus PhaseNetMatched Full3C	P	-0.1565	[-0.2033, -0.1123]
Stack minus PhaseNetMatched Z-only	P	-0.1699	[-0.2178, -0.1244]
Stack Full3C minus Z-only	S	+0.2850	[0.2293, 0.3418]
PhaseNetMatched Full3C minus Z-only	S	+0.2862	[0.2345, 0.3382]
Stack minus PhaseNetMatched Full3C	S	-0.0320	[-0.0720, 0.0066]
Stack minus PhaseNetMatched Z-only	S	-0.0309	[-0.0662, 0.0050]



502
503 Fig. 5. Stacker versus PhaseNet-style MATLAB baseline on the same 330 test records from
504 221 sources: (left) P and (right) S F1 at ± 100 ms, accepting detected and uncertain picks. Bars
505 show means. For stacking, each white point is a mean over three meta seeds for one base seed;
506 error bars are the sample SD of the three base-specific means. For the baseline, points are its
507 three independently initialized fits and error bars are their sample SD. These are seed SDs, not
508 confidence intervals; paired source-cluster intervals are reported in Table 7. Data and decoder
509 are shared, but inputs and compute budgets differ.

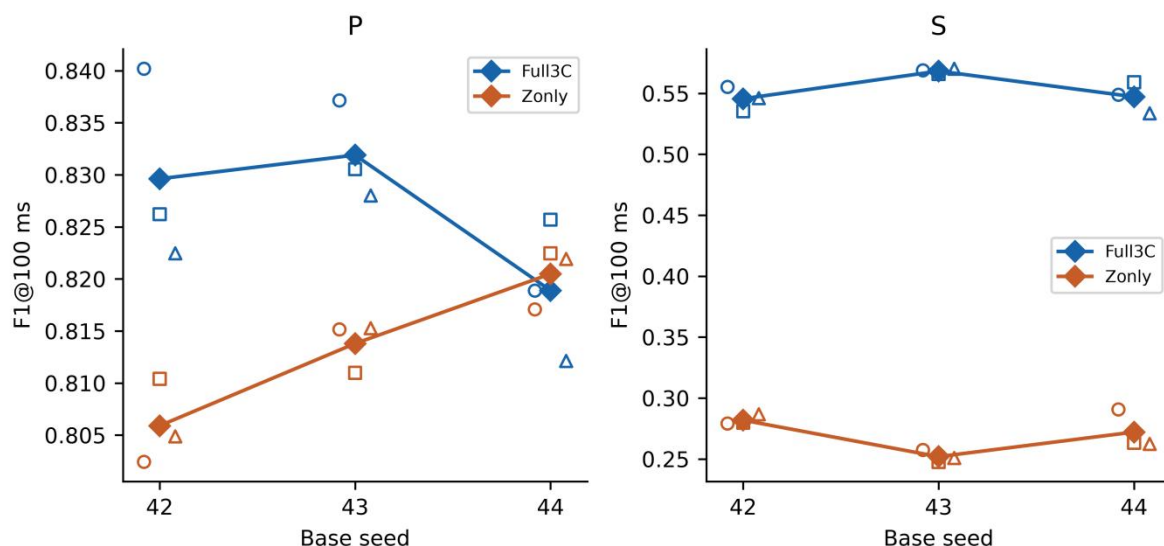


Fig. 6. Stacker initialization variability on one fixed source split: (left) P and (right) S F1 at ± 100 ms. Colour identifies input mode, open symbols identify individual meta seeds, and filled diamonds show the three-meta-seed mean within each base seed. Horizontal offsets separate overlapping observations; lines connect means only as a visual guide. Panel y-axis ranges differ to expose the small P variations. Meta fits sharing a base are dependent. The source partition, OOF assignments and augmentation seeds remain fixed; this figure does not measure split or geographic uncertainty.

Across the expanded grid, the conditional Full3C-minus-Z-only S contrast was positive for both approaches (Table 7). The stacker S difference was positive in every matched base/meta combination. Base-42-only contrasts are retained in Online Resource 1, Table S30.

For the expanded stacking average, the paired 95% interval for Full3C minus Z-only F1@100 ms was -0.0016 to 0.0295 for P and 0.2293 to 0.3418 for S. The P interval does not establish equivalence; the S interval supports a component-access advantage conditional on these models and test sources. PhaseNetMatched also showed an S component-access difference of 0.2862 (interval 0.2345 – 0.3382), indicating that the finding is not unique to stacking.

4.4. Retrospective sensitivity analyses

On 287 both-phase manual records from 188 sources, stacking S Full3C-minus-Z-only $\Delta F1$ was 0.2705 [0.2113, 0.3299], positive in all nine paired fits; PhaseNetMatched gave 0.2550 [0.2013, 0.3085]. This supports subset consistency, not annotation-error immunity. Both intervals included zero in the 20-record manual/no-historical-train-validation intersection, which still had historical test exposure (Online Resource 1, Table S25; Fig. S6).

Station stratification remains descriptive: the 12 records at nine currently unseen network-receiver keys provide only ten source clusters. Their S method-comparison intervals include zero and cannot establish station transfer (Online Resource 1, Table S26).

Across the predefined grid, accepted S Full3C-minus-Z-only $\Delta F1$ ranged from 0.2839–0.2856 for stacking and 0.2862–0.2873 for PhaseNetMatched; all pointwise intervals were positive. Quality invariance reduces accepted scoring to nine peak/window combinations. Detected-only ranges were 0.2855–0.3031 and 0.2828–0.2875. The baseline P advantage persisted throughout. Accepted S method intervals included zero throughout, whereas 18/27 detected-only Z-only S intervals favoured the baseline. Stacking P component intervals depended on scoring policy (Online Resource 1, Table S27; Fig. S7). These fixed-model diagnostics neither select a new operating point nor establish historical calibration provenance.

Base-42 ablations are secondary. Their exploratory S contrasts favor the full temporal representation over probability-only, waveform-only and no-dilation variants, but full-minus-logistic intervals include zero for both phases. These results do not isolate nonlinearity or establish a universal architecture benefit (Online Resource 1, Tables S5–S6 and S29–S30).

5. Discussion

5.1. Informatics implications

The checks separate artifact consistency, exclusion of specified dependencies within the corrected workflow, and independent predictive confirmation. Only the first two are established here. Waveform fingerprints connect exported records to archive entries, while membership and feature contracts connect those identities to fitted models and predictions.

553 This complements dataset interfaces such as SeisBench (Woollam et al. 2022) and the
554 entity/activity/derivation perspective of PROV-DM (Moreau and Missier 2013); it is not a
555 replacement for those tools or a claim of PROV conformance. For transfer, users must define a
556 scientifically appropriate grouping key and test which dependencies remain outside it (Roberts
557 et al. 2017).

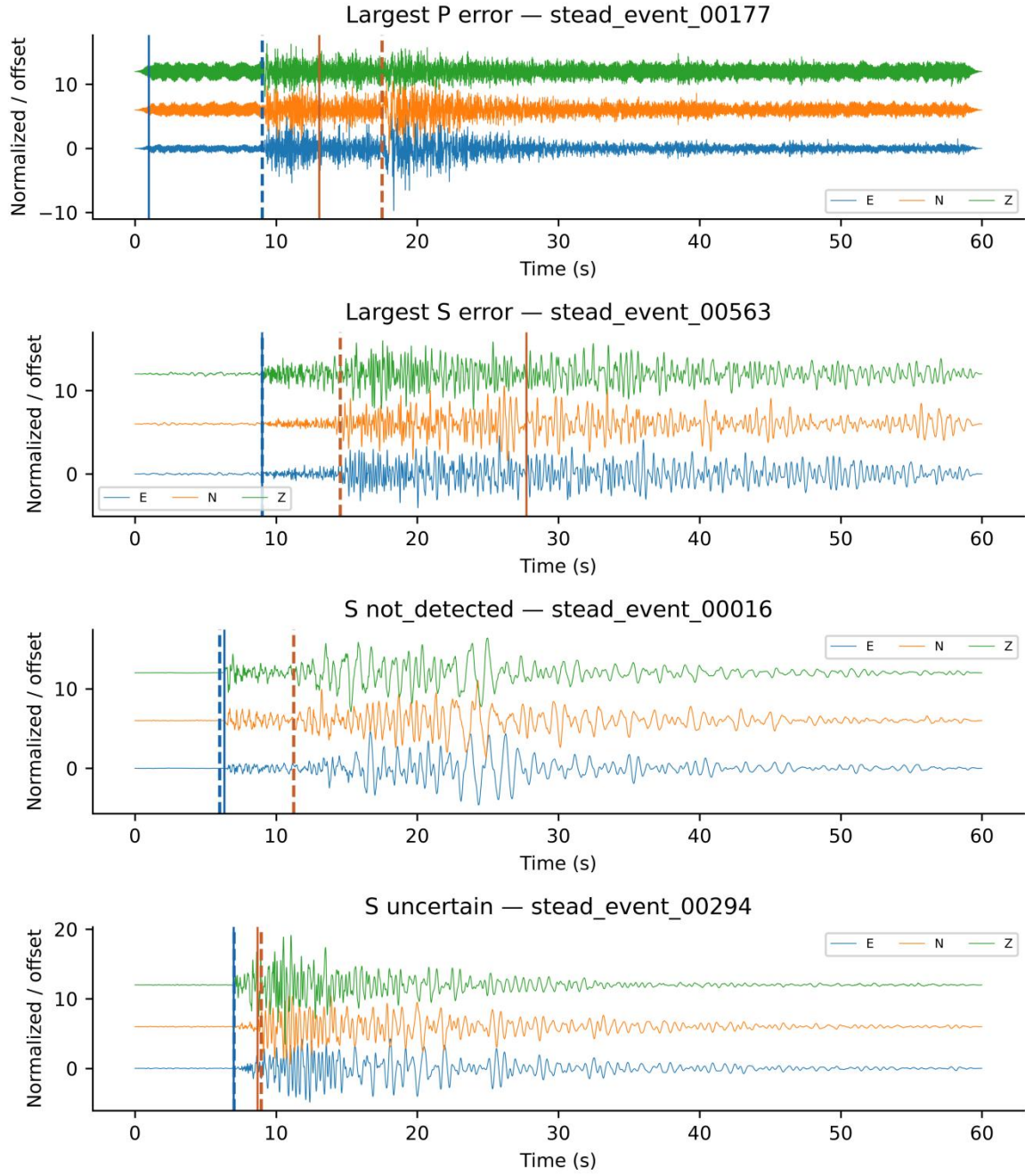
558 The practical design principle is to preserve identity at each interface rather than rely on
559 filenames or row order. A grouping manifest establishes what is excluded; a feature schema
560 establishes what is passed between learners; an artifact manifest establishes what was executed;
561 and replay tests the stored outputs. FAIR principles motivate reusable identifiers and metadata
562 (Wilkinson et al. 2016), but this implementation is not a FAIR certification or a general
563 provenance standard. No second-dataset portability benchmark was performed.

564 **5.2. Seismic interpretation**

565 The component-access difference is consistent with horizontal motion contributing useful
566 shear-wave information and persists across the three tested base seeds. It also includes changes
567 to the classical-base fallback and separately fitted learned bases, so it does not isolate any
568 individual channel. Source-split, station, and regional stability remain untested.

569 Constrained and unconstrained outputs are diagnostic of the implemented decoder, not an
570 isolated intervention on phase ordering: removing the S window also changes the quality-score
571 domain. Record-level outputs and decoder diagnostics accompany the frozen result package.
572 Ordered outputs do not show that the network learned a physical rule.

573 S timing remains a practical limitation: in the corrected Full3C base-42/meta-42 stacker,
574 accepted S RMSE was 1596.78 ms and 6.21% of accepted errors exceeded 1 s. High candidate
575 coverage is insufficient evidence for automatic catalogue use. PhaseNetMatched lowers mean
576 accepted-pick MAE, but neither workflow has been validated for continuous monitoring or
577 downstream location accuracy. Online Resource 1, Fig. S14 displays error tails; Fig. 7 shows
578 selected waveform cases.



579

580 Fig. 7. Selected Full3C base-42/meta-42 cases: largest accepted P error, largest accepted S
 581 error, and first S not-detected and uncertain cases when present. Components are normalized
 582 individually and offset. Dashed lines mark true arrivals; solid lines mark reported predictions.
 583 Event IDs identify the selection.

584 5.3. Limits of inference

585 The study uses one retrospectively recovered STEAD subset and fixed split. Initialization
 586 repetitions and source bootstrapping exclude partition and training uncertainty; base-42

587 ablations lack replication at other base seeds. Meta fits share OOF features, and outer
588 validation is reused for base selection, meta stopping and ensemble weighting. Most current
589 test sources had historical training/validation exposure. Manual-status sensitivity does not
590 quantify annotation error, and the unavailable historical selection recipe limits assessment of
591 selection bias. Twelve unseen-key records cannot establish station transfer. Threshold
592 sensitivity does not prove validation-only calibration; multiple intervals remain exploratory.
593 Regional transfer, continuous-stream false alarms, overlapping events and operational latency
594 were not evaluated. The regenerated CSVs restore byte-identical selected inputs in the tested
595 archive; this does not restore independent validation, prove another archive version equivalent,
596 or establish cross-hardware replay.

597 The additional comparator is a PhaseNet-style MATLAB adaptation, not the official PhaseNet
598 implementation or a pretrained benchmark. It uses the same data and decoder but differs in
599 architecture, input representation, minibatch size, and optimizer-step budget; the observed
600 differences cannot be assigned to one design choice. TwoStream-EQT and SeismoDual report
601 evaluation across additional datasets or acquisition settings (Peng et al. 2025; Tang and Chen
602 2026); their scores are not directly comparable here. The AIC score centre cannot exceed 9.99
603 s, whereas 234 of 330 test S arrivals and 15 test P arrivals occur later. This limits the classical
604 comparator and supplied features; the study is not a general comparison against optimized AIC
605 picking. These counts concern score centres before conditional decoding, not final error counts.

606 **5.4. Future directions**

607 Confirmation requires data outside prior research and development; another split of this
608 population cannot erase historical exposure. Additional source splits can measure partition
609 sensitivity, while nested validation separates selection stages. Independent datasets, matched
610 official picker implementations, continuous-record false alarms, latency and controlled
611 perturbations remain necessary tests of transfer and mechanism.

612 **6. Conclusions**

613 The study demonstrates an executable connection between waveform identity, grouped feature
614 generation, model provenance and reproducible evaluation. Unique matches in the local
615 STEAD archive define the retained records and sources; membership assertions and replay
616 make their computational use inspectable. Synthetic failures show what the implemented
617 checks reject. These outcomes establish artifact consistency and specified within-run
618 exclusions. Historical development exposure prevents an independent-confirmation claim, and
619 station, regional and independent-dataset generalization remain unestablished. A tested export
620 profile now regenerates all 2,234 retained CSV byte hashes from their mapped HDF5 traces,
621 while the original subset-selection history remains unavailable.

622 On this fixed split of 330 records from 221 sources held out from v2 fitting and validation,
623 accepted stacking P/S F1@100 ms was 0.8268/0.5536 for Full3C and 0.8134/0.2686 for Z-only.
624 Between-base SDs were 0.0069/0.0127 and 0.0073/0.0152. Full3C accepted-pick MAE was
625 166.62/610.29 ms, with coverage 99.06%/98.01%. Acceptance combines detected and
626 uncertain picks.

627 The S component difference was 0.2850 (conditional 95% interval 0.2293–0.3418).
628 PhaseNetMatched achieved P/S F1 of 0.9833/0.5857 (Full3C) and 0.9833/0.2995 (Z-only);
629 primary conditional intervals support its P advantage but include zero for both S method
630 differences. Base-42 ablations support exploratory S context/dilation gains without isolating
631 nonlinearity. Confirmation requires data kept outside prior development.

632 **Statements and Declarations**

633 **Author contributions**

634 **Rahmat Catur Wibowo:** Conceptualization, Methodology, Software, Validation, Formal
635 analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing,
636 Visualization, Supervision, Project administration. **Razki Alfatah Khairu Mahli:** Software,
637 Investigation, Validation, Visualization, Writing – review & editing. **Karyanto:** Software, Data

638 curation, Investigation, Writing – review & editing. **Hendra Amijaya:** Methodology,
639 Resources, Writing – review & editing. **Aristoteles:** Formal analysis, Validation, Writing –
640 review & editing. **Maman Hermana:** Validation, Writing – review & editing. **Muh Sarkowi:**
641 Conceptualization, Methodology, Supervision, Writing – review & editing.

642 **Code availability**

643 The software is MIT licensed. The implementation repository is
644 https://github.com/rcw3712/ICNN_MetaPicker_Curated_STEAD_MATLAB. Release source-
645 verified-v2.0.0, Git commit 8e82c781a1641191b6b56b15688d68e269b24918, is archived at
646 <https://doi.org/10.5281/zenodo.22824997>. Its source_verified_v2 directory contains the
647 recovery mapping, frozen partition and OOF protocol, training scripts, 106 checkpoints,
648 predictions, independent audits, inference receipts and threshold/station diagnostics, with an
649 English user guide, runnable synthetic examples and hash manifests. The downloaded archive
650 matches all 1,241 package-manifest entries. Earlier releases contain historical partitions and
651 must not be used to reproduce these corrected results. The archived release retains the earlier
652 manuscript title. The present ESI manuscript reorganizes the presentation and updates literature
653 and availability wording after archiving; fitted models, frozen predictions and reported
654 performance are unchanged. The subsequent revision adds a tested CSV-regeneration utility
655 and its receipts; these additions are not claimed to be in the existing DOI archive. The archive
656 is cited as Wibowo et al. (2026).

657 **Data availability**

658 The waveforms originate from STEAD (Mousavi et al. 2019), available through
659 <https://github.com/smousavi05/STEAD>. The retained 2,234 records map to 1,477 recovered
660 earthquake sources. Trace names, original source IDs and numerical waveform fingerprints
661 specify the selected observations independently of their local ordinal filenames. Raw STEAD
662 waveforms remain subject to the original distribution terms and are not redistributed here.
663 The retained trace mapping and waveform fingerprints identify the inputs. The accompanying
664 regenerate_retained_csv.py utility reconstructs the retained CSV representation from STEAD

665 HDF5 and rejects any file whose SHA-256 differs from the frozen CSV manifest. All 2,234
666 regenerated byte streams matched in the tested local archive. Raw waveforms are not bundled.
667 This recovers the selected exports, not the unavailable original selection/filtering decisions;
668 exhaustive uniqueness and reconstruction results apply to the tested archive version.
669 Historical-role and threshold/station diagnostics remain available.

670 **Competing interests**

671 The authors declare that they have no known competing financial interests or personal
672 relationships that could have appeared to influence the work reported in this paper. The
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679 **AI assistance in manuscript preparation**

680 During manuscript preparation, the authors used ChatGPT/Codex (OpenAI) and Claude
681 (Anthropic) for language editing, organization, and assistance drafting tables and captions.
682 After using these tools, the authors reviewed and edited the text, verified references and
683 consistency with the study results, and take full responsibility for the content of the publication.
684 AI assistance with the computational workflow is described in Section 3.

685 **Supplementary information**

686 Online Resource 1 (ESM_1.pdf) contains supplementary figures S1–S14 and tables S1–S31,
687 covering feature statistics, secondary base-42 results, hyperparameters, provenance, historical
688 roles, replay and sensitivity diagnostics. The accompanying reconstruction script and receipts
689 document the HDF5-to-CSV check.

690 **References**

691 Allen RV (1978) Automatic earthquake recognition and timing from single traces. Bulletin of
692 the Seismological Society of America 68:1521–1532.
693 <https://doi.org/10.1785/BSSA0680051521>

694 Baer M, Kradolfer U (1987) An automatic phase picker for local and teleseismic events.
 695 Bulletin of the Seismological Society of America 77:1437–1445.
 696 <https://doi.org/10.1785/BSSA0770041437>

697 Bai S, Kolter JZ, Koltun V (2018) An empirical evaluation of generic convolutional and
 698 recurrent networks for sequence modeling. arXiv preprint arXiv:1803.01271.
 699 <https://doi.org/10.48550/arXiv.1803.01271>

700 Cawley GC, Talbot NLC (2010) On over-fitting in model selection and subsequent selection
 701 bias in performance evaluation. Journal of Machine Learning Research 11:2079–2107.
 702 <https://www.jmlr.org/papers/v11/cawley10a.html>

703 Cheng A, Xu E, Yao P, Zhou Y, Huang S, Ye Z (2024) An automatic arrival time picking
 704 algorithm of P-wave based on adaptive characteristic function. Computers & Geosciences
 705 185:105544. <https://doi.org/10.1016/j.cageo.2024.105544>

706 Duerr RE, Downs RR, Tilmes C, Barkstrom B, Lenhardt WC, Glassy J, Bermudez LE,
 707 Slaughter P (2011) On the utility of identification schemes for digital earth science data: an
 708 assessment and recommendations. Earth Science Informatics 4:139–160.
 709 <https://doi.org/10.1007/s12145-011-0083-6>

710 Ioffe S, Szegedy C (2015) Batch normalization: accelerating deep network training by
 711 reducing internal covariate shift. Proceedings of the 32nd International Conference on Machine
 712 Learning, PMLR 37:448–456. <https://doi.org/10.48550/arXiv.1502.03167>

713 Jover-Alfaro J, Arias-Antúnez E, Mateo-Cortés JA (2026) Forecasting earthquakes by Machine
 714 Learning techniques: lights and shadows. Earth Science Informatics 19:24.
 715 <https://doi.org/10.1007/s12145-026-02078-x>

716 Kapoor S, Narayanan A (2023) Leakage and the reproducibility crisis in machine-learning-
 717 based science. Patterns 4:100804. <https://doi.org/10.1016/j.patter.2023.100804>

718 Kingma DP, Ba J (2015) Adam: a method for stochastic optimization. Proceedings of the 3rd
 719 International Conference on Learning Representations (ICLR 2015).
 720 <https://doi.org/10.48550/arXiv.1412.6980>

721 Maeda N (1985) A method for reading and checking phase times in auto-processing system of
 722 seismic wave data. Zisin (Journal of the Seismological Society of Japan, Second Series)
 723 38:365–379. https://doi.org/10.4294/zisin1948.38.3_365

724 Moreau L, Missier P (eds) (2013) PROV-DM: The PROV Data Model. W3C Recommendation,
725 30 April 2013. <https://www.w3.org/TR/2013/REC-prov-dm-20130430/>

726 Mousavi SM, Ellsworth WL, Zhu W, Chuang LY, Beroza GC (2020) Earthquake transformer—
727 An attentive deep-learning model for simultaneous earthquake detection and phase picking.
728 *Nature Communications* 11:3952. <https://doi.org/10.1038/s41467-020-17591-w>

729 Mousavi SM, Sheng Y, Zhu W, Beroza GC (2019) STanford EArthquake Dataset (STEAD): A
730 global data set of seismic signals for AI. *IEEE Access* 7:179464–179476.
731 <https://doi.org/10.1109/ACCESS.2019.2947848>

732 Münchmeyer J, Woollam J, Rietbrock A, Tilmann F, Lange D, Bornstein T, Diehl T, Giunchi C,
733 Haslinger F, Jozinović D, Michelini A, Saul J, Soto H (2022) Which picker fits my data? A
734 quantitative evaluation of deep learning based seismic pickers. *Journal of Geophysical*
735 *Research: Solid Earth* 127:e2021JB023499. <https://doi.org/10.1029/2021JB023499>

736 Peng L, Li L, Mousavi SM, Zeng X, Beroza GC (2025) TwoStream-EQT: A microseismic
737 phase picking model combining time and frequency domain inputs. *Computers & Geosciences*
738 204:105991. <https://doi.org/10.1016/j.cageo.2025.105991>

739 Roberts DR, Bahn V, Ciuti S, Boyce MS, Elith J, Guillera-Arroita G, Hauenstein S,
740 Lahoz-Monfort JJ, Schröder B, Thuiller W, Warton DI, Wintle BA, Hartig F, Dormann CF
741 (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic
742 structure. *Ecography* 40:913-929. <https://doi.org/10.1111/ecog.02881>

743 Sleeman R, van Eck T (1999) Robust automatic P-phase picking: An on-line implementation in
744 the analysis of broadband seismogram recordings. *Physics of the Earth and Planetary Interiors*
745 113:265–275. [https://doi.org/10.1016/S0031-9201\(99\)00007-2](https://doi.org/10.1016/S0031-9201(99)00007-2)

746 Srivastava N, Hinton G, Krizhevsky A, Sutskever I, Salakhutdinov R (2014) Dropout: a simple
747 way to prevent neural networks from overfitting. *Journal of Machine Learning Research*
748 15:1929–1958. <https://www.jmlr.org/papers/v15/srivastava14a.html>

749 Tang K-W, Chen K-Y (2026) SeismoDual: A dual-domain deep learning framework for robust
750 seismic phase picking. *Computers & Geosciences* 207:106080.
751 <https://doi.org/10.1016/j.cageo.2025.106080>

752 Vargas-Zamudio KA, Perton M, Legrand D, Sánchez-Sesma FJ (2026) Physically driven
753 feature engineering for deep learning applications in seismo-volcanic signal analysis. *Earth*
754 *Science Informatics* 19:11. <https://doi.org/10.1007/s12145-025-02062-x>

755 Wibowo RC, Mahli RAK, Karyanto, Amijaya H, Aristoteles, Hermana M, Sarkowi M (2026)
 756 Source-Grouped Evaluation of Temporal Stacking for Seismic Phase Picking with Three-
 757 Component Data. Version source-verified-v2.0.0. Zenodo.
 758 <https://doi.org/10.5281/zenodo.22824997>

759 Wilkinson MD, Dumontier M, Aalbersberg IJ, Appleton G, Axton M, Baak A, Blomberg N,
 760 Boiten JW, da Silva Santos LB, Bourne PE, Bouwman J, Brookes AJ, Clark T, Crosas M, Dillo
 761 I, Dumon O, Edmunds S, Evelo CT, Finkers R, Gonzalez-Beltran A, Gray AJG, Groth P, Goble
 762 C, Grethe JS, Heringa J, 't Hoen PAC, Hooft R, Kuhn T, Kok R, Kok J, Lusher SJ, Martone
 763 ME, Mons A, Packer AL, Persson B, Rocca-Serra P, Roos M, van Schaik R, Sansone SA,
 764 Schultes E, Sengstag T, Slater T, Strawn G, Swertz MA, Thompson M, van der Lei J, van
 765 Mulligen E, Velterop J, Waagmeester A, Wittenburg P, Wolstencroft K, Zhao J, Mons B (2016)
 766 The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*
 767 3:160018. <https://doi.org/10.1038/sdata.2016.18>

768 Wolpert DH (1992) Stacked generalization. *Neural Networks* 5:241–259.
 769 [https://doi.org/10.1016/S0893-6080\(05\)80023-1](https://doi.org/10.1016/S0893-6080(05)80023-1)

770 Woollam J, Münchmeyer J, Tilmann F, Rietbrock A, Lange D, Bornstein T, Diehl T, Giunchi C,
 771 Haslinger F, Jozinović D, Michelini A, Saul J, Soto H (2022) SeisBench—A toolbox for
 772 machine learning in seismology. *Seismological Research Letters* 93:1695–1709.
 773 <https://doi.org/10.1785/0220210324>

774 Zhu W, Beroza GC (2019) PhaseNet: A deep-neural-network-based seismic arrival-time
 775 picking method. *Geophysical Journal International* 216:261–273.
 776 <https://doi.org/10.1093/gji/ggy423>